

MAGNETOM

Troubleshooting Guide System

Cooling

Avanto
Espree

Document Version

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1	Strategy	4
1.1	General	4
1.1.1	Error messages	4
1.1.2	SEP	5
1.1.3	Chiller	5
2	Procedures	9
2.1	Drain procedure of the ACS	9
2.1.1	Required Tools	9
2.1.2	Procedure	9
2.2	Filling from the ACS	12
2.2.1	Required Tools	12
2.2.2	Procedure	13
2.3	SEP	19
2.3.1	Controller status	19
2.4	Drain procedure of the SEP (secondary water circuit)	20
2.4.1	Required Tools	20
2.4.2	Procedure	21
2.5	Filling from the SEP	26
2.6	Testing the expansion vessel	30
2.6.1	Required tools	30
2.6.2	Procedure	30
2.6.3	Pressurizing the expansion vessel	32
2.6.4	Concluding steps	32
2.7	Drain procedure of the IFP	34
2.7.1	IFP with old type of strainer	34
2.7.2	Draining procedure for IFP with new type of strainer	39
2.8	Chiller KCC 215-L-U/S, -S	40
2.8.1	Chiller interface device	40
2.8.2	Switch off / on procedure of the Chiller	43
2.8.3	Automatic restart procedure for the chiller	43
2.8.4	Drain procedure of the chiller (water circuit)	44
2.8.5	Filling from the chiller	46
2.8.6	Bleeding the Chiller water pump	51
2.9	Testing the ethylene glycol level	54
2.9.1	Required Tools	54
2.9.2	Procedure	54
2.10	Testing the water flow to the RFPA and the gradient power stage	55
2.10.1	Required tool	55
2.10.2	Test procedure of the RFPA	55
2.10.3	Test procedure for the gradient power amplifier	58
2.11	Conversion list of temperature sensors	59
3	Changes to previous version	60
4	List of Hazard IDs	61

1.1 General

In this section, you will find the **strategy** and **test procedures** for troubleshooting the Cooling.

The main tool for troubleshooting consists of error messages delivered from the ACS, IFP, SEP and the Chiller and shown in the error log file of the service platform.

1.1.1 Error messages

Refer to "Cooling Error Messages" (MR00-000.840.18.xx.02) (→ Cooling Error Messages VB17 / MR00-000.840.18).

1.1.2 SEP

Symptom	Area of Trouble (Components/function)	Probable Cause	Service action
Secondary water temperature is out of specification	Actuator	Cable connection	Check cable connection at the actuator Y1, Y2. Check cable connection between actuator and controller. Perform a reset.

1.1.3 Chiller

Symptom	Area of Trouble (Components/function)	Probable Cause	Service action
Malfunction of Chiller	Power supply	Power fault asymmetry Overvoltage or low voltage	Check power voltage line and asymmetry relay 5A1.
		Temperature sensor defective	Check sensor 9B1: » clamp sensor » measure the resistance
		Simatic 8N1, 8A1, 8A2 and 8A3	Check MMC-Card: » Check power supply 24 VDC 7U1 » Check fuses 7F1 and 7F2 Contact the chiller company
		Chiller "not ok"	Check fuse 28F1 Check data transfer cable Check overvoltage protection

Symptom	Area of Trouble (Components/function)	Probable Cause	Service action
Malfunction of pump 21M1	Water circuit	Main switch 5Q1 not switched on	Switch on the main switch 5Q1
		Control switch 14S1 on "OFF"	Switch control switch 14S1 to "AUTO"
		Main fuse 21Q1 defective	Replace fuse 21Q1
		Fuse for control current defective	Replace fuse
		Pump Motor 5M1 defective	Replace pump 21M1
		Flow switch 11B1 responded	Check filling water pressure Check water flow
		Shortage of water	Check filling water pressure Clean strainer
		Overload trip 21Q1 of pump protection interrupted control circuit	Main switch to "0"; push in overload trip
Pump makes gurgling noise	Water circuit	Water circuit not completely vented	Vent and fill up with ethylene glycol water mixture.
Compressor stops	Compressor 22M1 and 25M1 stops	Klixon/INT69 tripped 22A1 and 25A1	Check fuses 22Q1 and 25Q1 wait until compressor cooled down ; clean condenser or provide fresh air supply Contact the chiller company
		Klixon/INT69 22A1 and 25A1 defective	Contact the chiller company

Symptom	Area of Trouble (Components/function)	Probable Cause	Service action
Malfunction of refrigerating machine	refrigerant circuit	Simatic stopped machine Return temperature too cold	Contact the chiller company
		Low pressure in refrigerant circuit - Plant loses refrigerant - dryer in liquid pipe dirty - pressure relief valve defective - Solenoid valves 23Y1 and 26Y1 in liquid pipe defective	Contact the chiller company
		High pressure in refrigerant circuit - condenser dirty - fan defective - outside temperature too high - pressure sensor 24B1 and 27B1 for condenser control defective - frequency inverter defective	Contact the chiller company Check fuse 24Q1 and 27Q1 Spray condenser with water Contact the chiller company replace frequency inverter 24U1 and 27U1
Refrigerating machine starts and stops short term	Refrigerant circuit	not enough fresh air supply for condenser. High-pressure pressostat tries to protect refrigerating machine against overload	Provide enough fresh air supply and fresh air removal. Get rid of short-circuit across fresh air and exhaust air. Contact the chiller company
		Not enough pressure in refrigerant circuit. Part of the refrigerant escaped, diminished pressure switch, shut down compressor	Contact the chiller company
Not enough refrigeration power	Water circuit	Air in water circuit	Vent system Refill water circuit

Symptom	Area of Trouble (Components/function)	Probable Cause	Service action
	Refrigerant circuit	Not enough fresh air supply for condenser	Provide enough fresh air supply and fresh air removal. Get rid of short-circuit across fresh air and exhaust air. Contact the chiller company
	Refrigerant circuit	Not enough refrigerant in circuit	Contact the chiller company
Power fault » automatic restart	n.a.	n.a.	refer to:(→ Automatic restart procedure for the chiller / Page 43)
Remarks: Check the eventlog of the MR system for errors and warnings generated by the Magnet Supervision.			

2.1 Drain procedure of the ACS



Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

2.1.1 Required Tools

- Filling adapter, delivered with the system.
- Bucket
- Water hose, delivered with the system.

2.1.2 Procedure

- Close the valves on top of the ACS.

Fig. 1: ACS shutoff valve



(1) Open-end spanner

- Remove the quick connections of the water hoses from the GPA.

Fig. 2: ACS water filling



- (1) Return circuit
- (2) Supply circuit
- (3) Filling pipe

- Connect water hose with filling adapter fitted to the return circuit valve.

Fig. 3: Filling port



(1) Filling port valve

- Position the bucket in front of the cooler and put the cooler drain hose in it (the ACS holds approximately 8 l of water).
- Open the valve of the ACS return circuit port.
- Connect the water hose with the filling adapter fitted to the supply circuit valve.
- Open the valve of the ACS supply circuit valve.
- Vent ACS. Loosen the screws at the connection on top of the cabinet. Air entering the system will help drain the ACS.

2.2 Filling from the ACS

MR-system with IFP can be filled from the ACS or the chiller.

MR-system with SEP should be filled from the SEP.



System with IFP.

Filling the system via the return circuit, low-pressure circuit if the chiller is on.

The water circuit from the chiller to the MR system needs to be filled with 35% to 38% ethylene glycol.

35% to 38% vol. ethylene glycol mixture corresponds to approx. -21 °C to -23 °C (-5.8 °F to -9.4 °F).

Only ethylene glycol may be used.

When starting up the chiller, the chiller start-up personnel should leave 30 liters of ethylene glycol mixture at the system.

This reserve supply can be used for refills.



Do not use any connection to a drinking water system!

2.2.1 Required Tools

Fig. 4: Filling adapter



- Water pump (Hand pump or water pump with drill)
- Filling adapter.
- Water hose
- Check that the valve of the filling port of the return circuit is closed.

2.2.2 Procedure

- Fit water hose with filling adapter.
Filling adapter and water hose are delivered with the system.

Fig. 5: ACS water filling



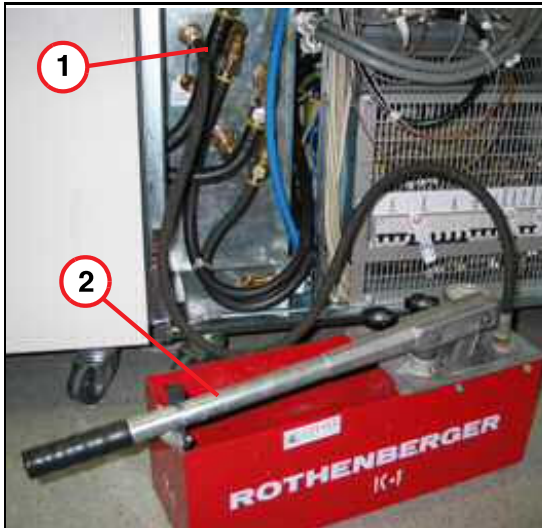
- (1) Return circuit
(2) Supply circuit

Fig. 6: Fill and drain port



- Remove the protective cap from the return circuit filling port.

Fig. 7: Hand pump



- Connect the water hose to a hand pump or a water pump with drill.
 - » **Do not use any connection to a drinking water system!**

- (1) Fill valve
(2) Hand pump

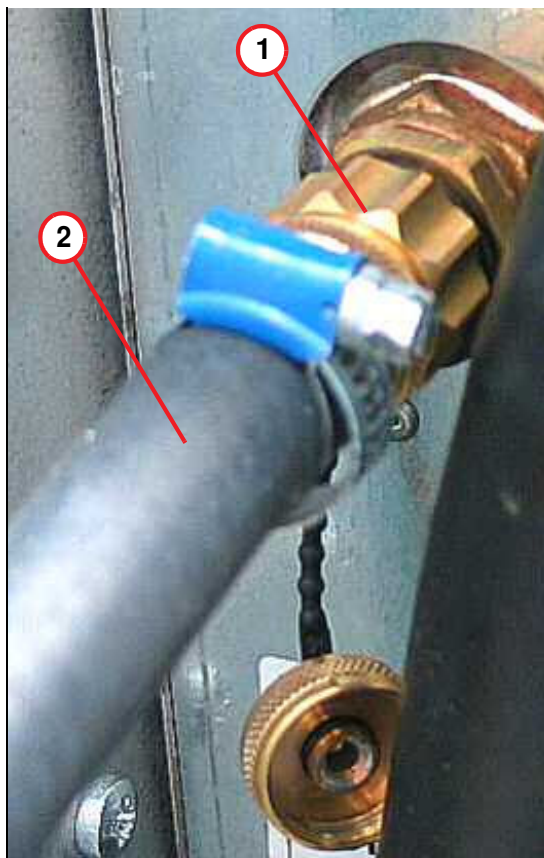
Fig. 8: Water pump with drill



- (1) Drill machine
- (2) Water pump

- Fill the pump with an ethylene glycol mixture.

Fig. 9: Filling port



- (1) Filling port valve
- (2) Water hose

- Connect the other end of the water hose to the fill port of the return circuit.



Before connecting the hose to the ACS, the hose should be filled with water and not contain air.

Fig. 10: ACS water filling



- (1) Return circuit
- (2) Supply circuit
- (3) Filling pipe

Fig. 11: Filling port



(1) Filling port valve

- Open the valve of the ACS fill port.
 - Start pumping
 - Close the fill port.
 - Open the SESO platform "Magnet & Cooling": Calibrate the flow sensors.
- Check the pressure:

System with Chiller - IFP

Magnet & Cooling / Status

- Check water pressure, supply
 - Check water pressure, return
- » Calculate the mathematical difference between the water pressure supply and the water pressure return (water pressure supply minus water pressure return).
- » The difference must be in the range between 4.0 bar and 4.5 bar.
- » Refill if water pressure return shows <0.5 bar or pressure difference is out of specification.
- » Check warning level of "Water Pressure Return":
- Chiller and pipe routing on the same level or
Chiller is below MR system level
Warning level is 0.4 bar.
- Chiller is above MR system level or
Chiller and MR system are on the same level, but the pipe routing is above the MR system. Use the following formula:
- warning level = 0.1 bar (over pressure) + 0.1 bar + 0.01 bar/m (flow resistance) + 0.1 bar/m (difference of level)
- e.g. chiller 10 m above the Mr system, pipe length 20 m: Warning level = 0.1 bar + 0.01 bar/m * 20 m + 0.1 bar/m * 10 m warning level = 1.3 bar

- Disconnect the water hose
- Place the protective cap on the fill port.

2.3 SEP

2.3.1 Controller status

LED	Normal	Description
L1	blinks	program active
L2	Off	service mode S4
L5	*	communication o.k.
L7	Off On	no fault Error indication
L12	On	50/60 Hz
L13	On	fill pressure
L14	On	No overload pump
L15	On	No overload MREF
L16	*	MCU reset
L17	*	mixer cooler
L18	*	mixer warmer
L19	*	cooling pump

* on/off depends on actual controller status.

Switch / Button	Description
S1	Press S1 button to switch off the water pump and the helium compressor. Water pump and helium compressor are switching back on after approx. 2 minutes.
S4	Auto/service switch

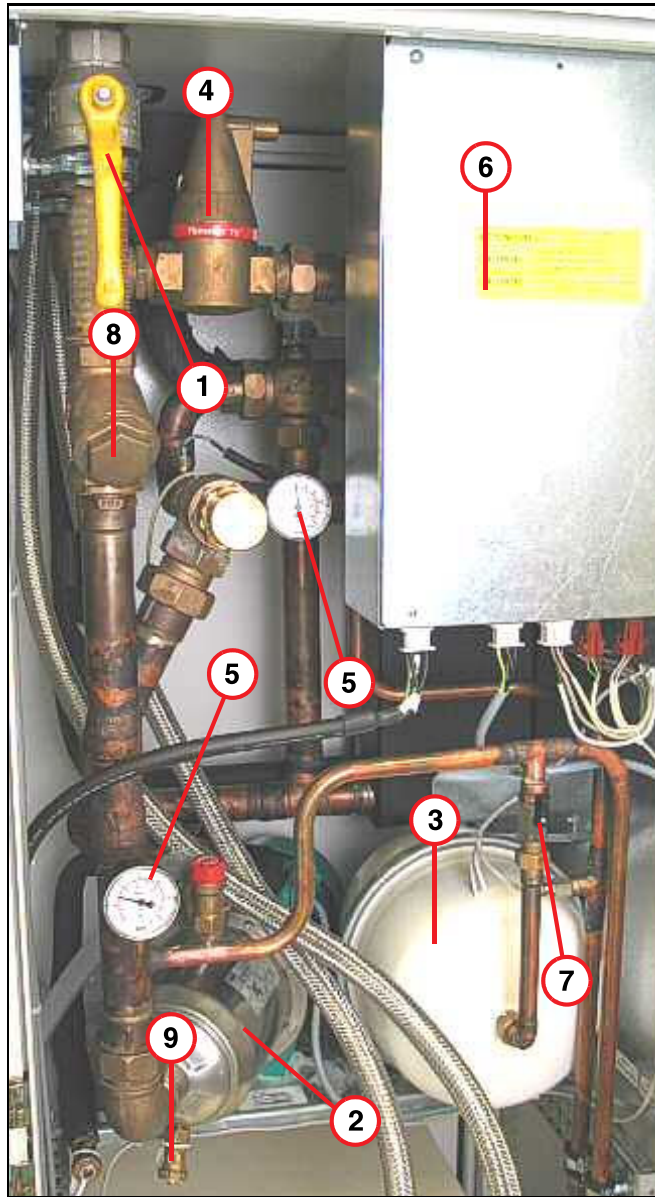
2.4 Drain procedure of the SEP (secondary water circuit)

2.4.1 Required Tools

- Filling adapter
- Bucket
- Water hose

2.4.2 Procedure

Fig. 12: SEP



- (1) Shut-off valve
- (2) Water pump
- (3) Expansion chamber
- (4) Air bleeder
- (5) Manometer
- (6) Electronics box
- (7) Valve
- (8) Dirt trap
- (9) Fill and drain valve

- Switch OFF the on-site line voltage to the SEP, F102.
- Close the shutoff valves of the SEP.

Fig. 13: Shut-off valve



- Close the valve of the expansion chamber

Fig. 14: Expansion chamber



(1) Valve

- Connect the water hose to the filling and drain port and run the hose to a drain or a container.

Fig. 15: Fill and drain valve



Fig. 16: Fill- and drain valve



- Open the drain port using the cap of the drain valve.
- Drain the water circuit. Volume of the water circuit to be drained is about 15 liters.
- Loosen the screw of the dirt trap a little. Air entering the system will help drain the SEP

2.5 Filling from the SEP

Fig. 17: Water filling



- Switch off F102 on top of ACC
- Connect the water hose to industrial or process water, or use a hand pump, or pump with a drill.



Do not use any connection to a drinking water system!

- Connect the water hose from the on-site water supply to the filling port of the SEP



Before connecting the hose to the SEP, the hose should be filled with water and not contain air.

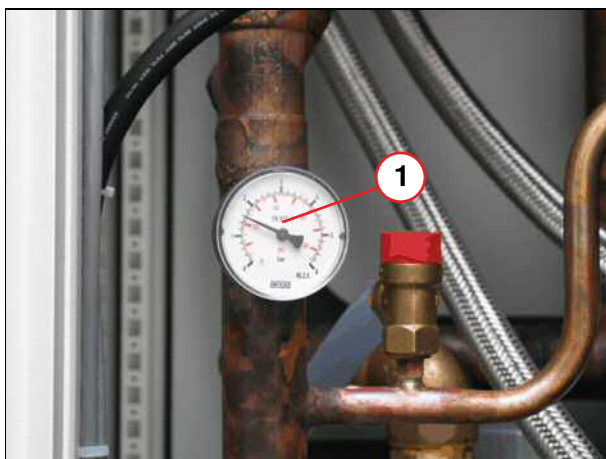
Fig. 18: Fill and drain valve



- Open the valve using the back of the cap
- Close the valve when the pressure manometer indicates:

SEP water pump off	Pressure manometer: Pressure 1.0 $\pm$ 0.5 bar
--------------------	---

Fig. 19: SEP pressure manometer



(1) Low pressure (pressure manometer)

- Switch on the SEP and set the water pressure.

Setting the water pressure

1. After the pump has started, the water pressure will drop since the air in the circuit escapes through the ventilation valve.
2. If the pressure drops below **0.5 bar**:
 - Switch off water pump with F3.
 - Fill the water circuit to 1.0 - 0.5 bar.
 - Switch on the water pump with F3.
3. The procedure has to be repeated if the water pressure drops below **0.5 bar**.
4. Calculate the mathematical difference between the high pressure (P_{out}) and low pressure (P_{in}).
 - » This difference (P_{out} minus P_{in}) must be at 50 Hz: 3.0 ± 0.3 bar
 - » This difference (P_{out} minus P_{in}) must be at 60 Hz: 3.8 ± 0.3 bar

The pressure difference can be adjusted with the bypass valve.

5. If the pressure does not drop below **0.5 bar** within 30 minutes, end the filling procedure for the SEP. Remove the hose and install the protection cup again.
- Open the SESO platform "Magnet&Cooling"
 - Calibrate the flow sensors.
 - Check the water pressure via "Magnet&Cooling"

System with SEP

Open the Status platform

Read out the water pressure with the water pump running:

- » ACS water pressure, supply
- » ACS water pressure, return
- » Calculate the mathematical difference between water pressure high and water pressure low (water pressure supply minus water pressure return)

System with SEP

Open the Status platform

- » Pressure difference (water pressure supply minus water pressure return) at 50 Hz:
3.0 bar +/-0.3 bar
- » Pressure difference (water pressure supply minus water pressure return) at 60 Hz:
3.8 bar +/-0.3 bar

Refill if ACS water pressure, return < 1.3 bar or pressure difference is out of specification.
Warning level for pressure return is at 0.8 bar

Warning level system with SEP

Water pressure, supply max.	6.0 bar
Water pressure, return min.	0.8 bar
Water pressure, return max.	2.8 bar
Water temperature, supply max.	25C

2.6 Testing the expansion vessel



Before replacing the pump due to pressure problems, test the pressure of the expansion vessel.

2.6.1 Required tools

- Air pump with autovalve.(→ Fig. 20 Page 30)
- Measuring device part number 3055514
- Standard tool set

Fig. 20: Air pump



Fig. 21: Autovalve



2.6.2 Procedure

- Switch off the line voltage to the SEP, F102.

Fig. 22: Shut-off valve



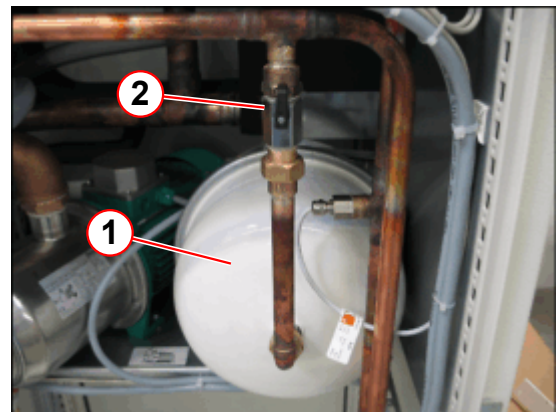
- Close the secondary water circuit's shutoff valves (inlet and outlet) of the.

- Drain the secondary water circuit to atmospheric pressure.
 - » The shutoff valve of the expansion vessel must be open. (→ 2/ Fig. 24 Page 31)

Fig. 23: Manometer



Fig. 24: Expansion vessel



- (1) Expansion vessel
(2) Valve

- Remove the protective cover on the rear side of the expansion vessel. (→ Fig. 25 Page 32)
- Remove the cap from the valve. (→ Fig. 26 Page 32)

Fig. 25: Protective cover



Fig. 26: Cap



- Connect the measuring device to the fill valve.(→ Fig. 27 Page 32)
- Check the pressure.
 - » Pressure should be 0.5 bar.
 If not, pressurize the vessel with the air pump.

Fig. 27: Measuring device



Fig. 28: Measuring device



2.6.3 Pressurizing the expansion vessel

- Connect the air pump to the fill valve.
- Pressurize the vessel to 0.5 bar.

2.6.4 Concluding steps

- Undo the cap and the protective cover on the fill valve.
- Fill the secondary water circuit with water.
- Open the shutoff valves.
- Switch on F102.
- Start the water pump.

- Start the compressor.
- Check the water pressure after the air is bled.

2.7 Drain procedure of the IFP



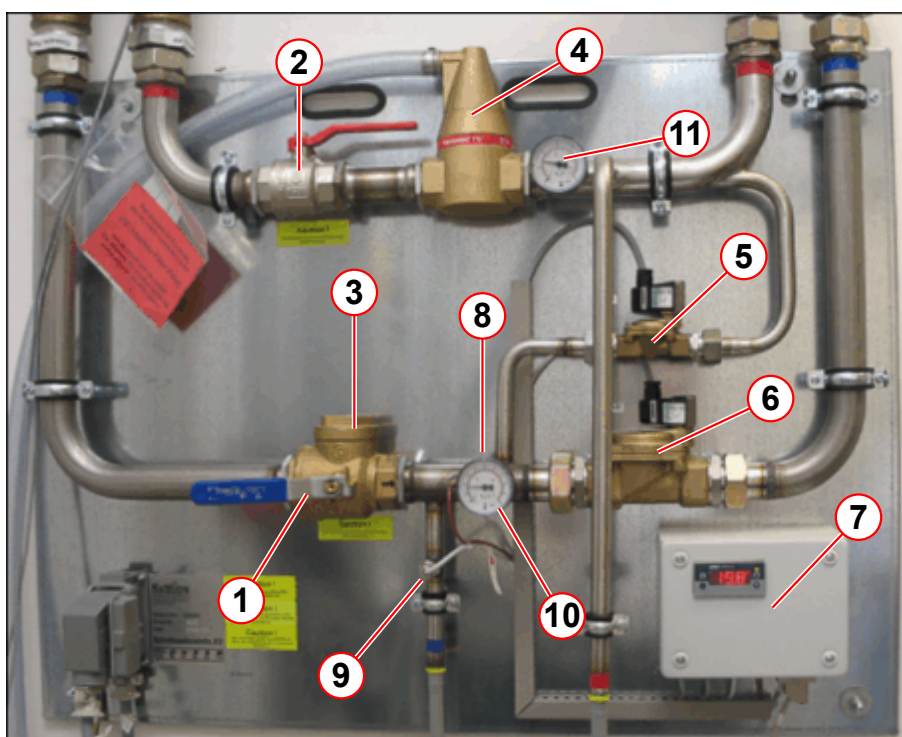
Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

2.7.1 IFP with old type of strainer



This procedure is not possible on the IFP with the new strainer shown in (→ 1,3/ Fig. 29 Page 34)

Fig. 29: Interface and filter panel (IFP)



- (1) Valve 1
- (2) Valve 2
- (3) Strainer
- (4) Bleeder
- (5) Bypass valve
- (6) Straight-through valve
- (7) Temperature controller
- (8) Temperature sensor
- (9) Flow sensor
- (10) Water pressure, supply
- (11) Water pressure, return

2.7.1.1 Required tools

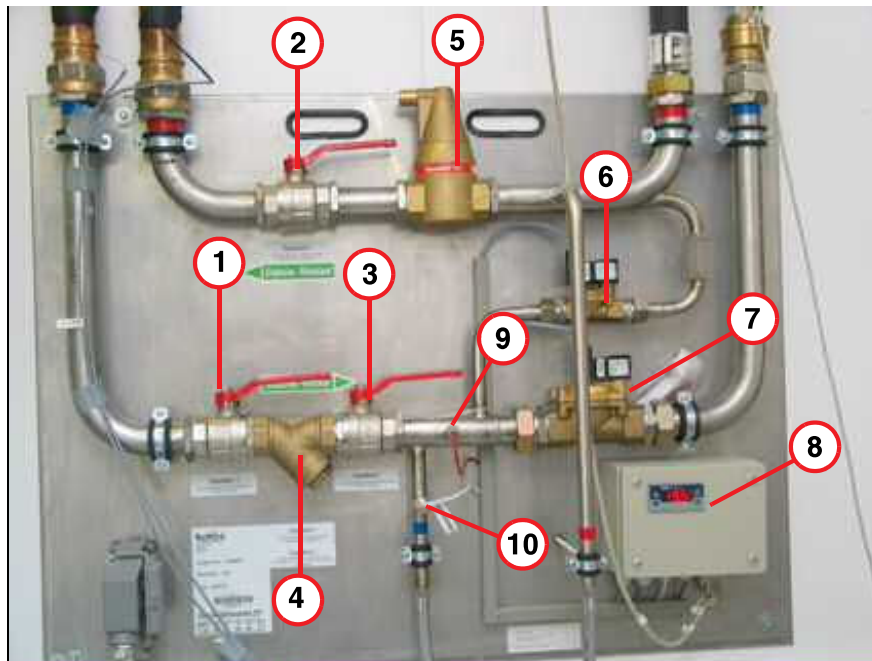
- Adjustable open-end spanner
- Bucket

- SW 60 open-end spanner

2.7.1.2 Procedure

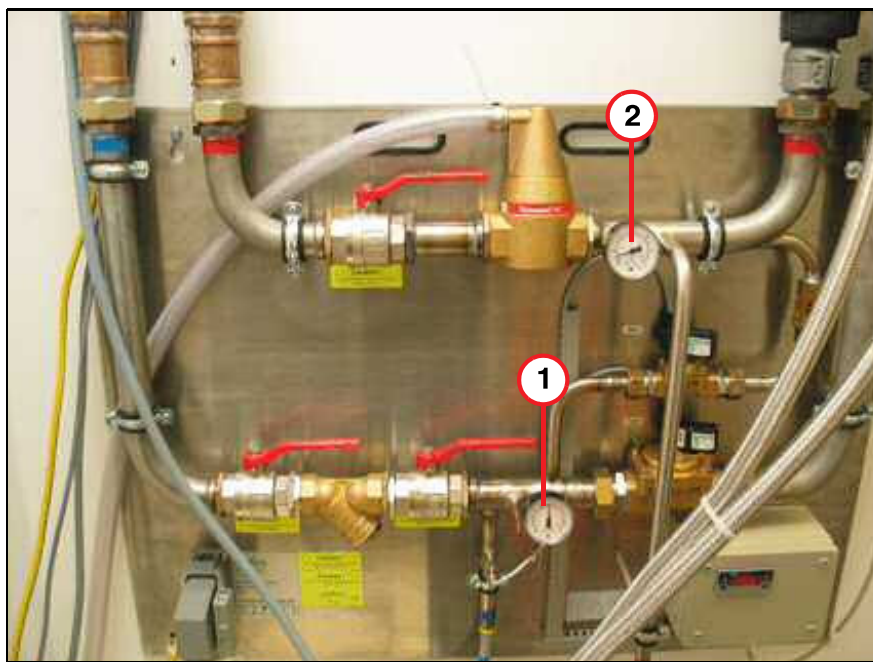
- Switch off the chiller
- Switch off F1 on the ACS, straight-through valve closes
- Close valve 1, 2, and 3 on the IFP

Fig. 30: Interface and filter panel (IFP)



- (1) Valve 1
- (2) Valve 2
- (3) Valve 3
- (4) Strainer
- (5) Bleeder
- (6) Bypass valve
- (7) Straight-through valve
- (8) Temperature control
- (9) Temperature sensor
- (10) Flow sensor

Fig. 31: IFP with pressure manometer



- (1) Water pressure supply
- (2) Water pressure return

- Close "to SEP/IFP" valve and "from SEP/IFP" valve on the ACS.

Fig. 32: ACS shutoff valve



(1) Open-end spanner

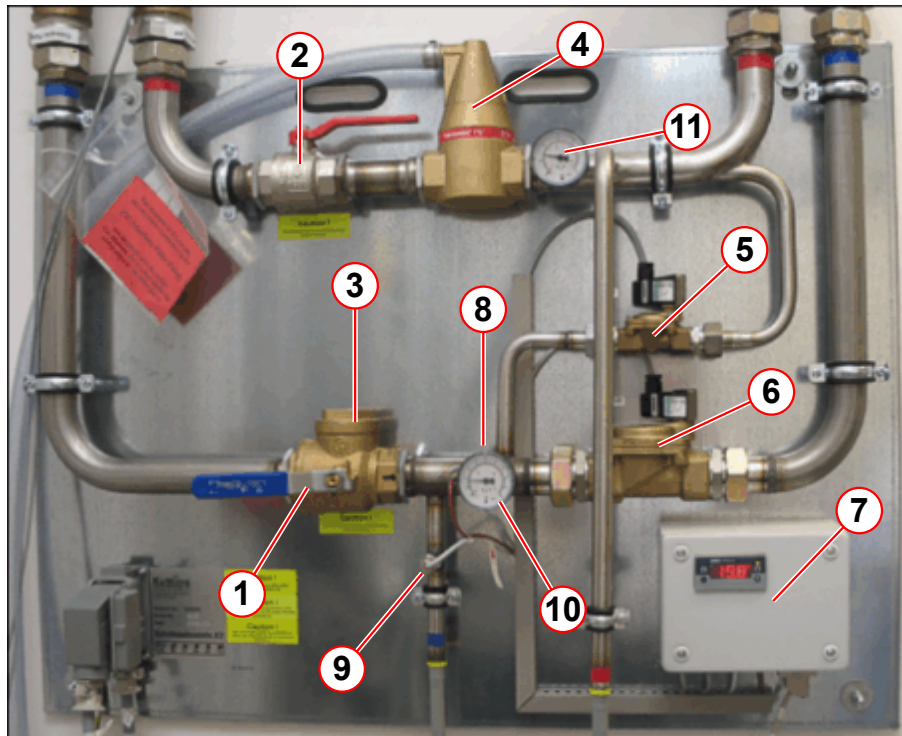
Fig. 33: Strainer



- Open the strainer. Collect the water in the bucket
- Open valve 3 of the IFP slowly to drain the IFP. (→ 3/ Fig. 30 Page 35)
- Loosen the upper part of the bleed-er, or another union end.
The air entering the system will help drain the IFP.

2.7.2 Draining procedure for IFP with new type of strainer

Fig. 34: Interface and filter panel (IFP)



- (1) Valve 1
- (2) Valve 2
- (3) Strainer
- (4) Bleeder
- (5) Bypass valve
- (6) Straight-through valve
- (7) Temperature controller
- (8) Temperature sensor
- (9) Flow sensor
- (10) Water pressure, supply
- (11) Water pressure, return

Procedure

- Switch off helium compressor.
- Switch off the chiller.
- Close valve 1 and valve 2 of the IFP.
- Close "to SEP/IFP" valve and "from SEP/IFP" valve on the ACS.
- Close the valve on the water pipes at the compressor.
- Unscrew the water hose from the compressor.
 - » Water will run out from the compressor!
- Run the hose to a container and open the valves to drain the IFP.

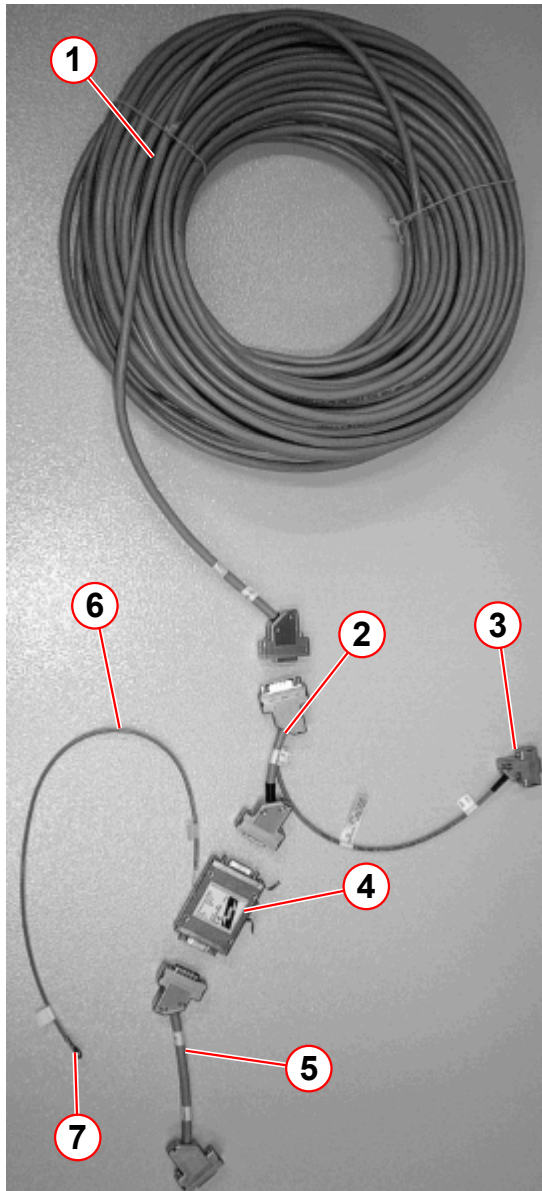
2.8 Chiller KCC 215-L-U/S, -S

2.8.1 Chiller interface device



To connect the chiller signal to the ACC cabinet, use only the complete set containing 2 cables (long and short cable), a grounding cable, and the overvoltage protection device.

Fig. 35: Chiller interface kit



- (1) Interface cable
- (2) Y-cable
- (3) ACC X96
- (4) Overvoltage protector
- (5) Cable Filter - ACC - X81
- (6) Ground wire
- (7) Filter box earth connector



If the signal cable is connected to the ACC cabinet without the overvoltage protection device and the short interface cable the relay 19K1 will be destroyed.

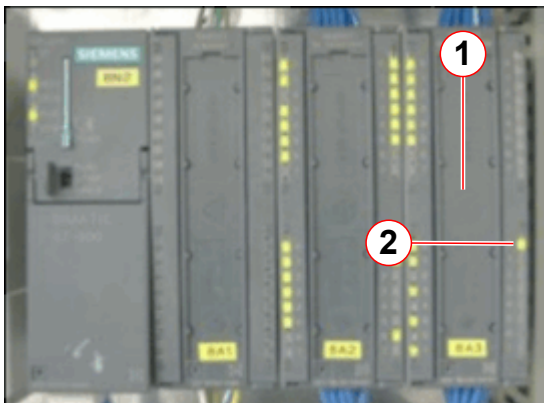
The relay 19K1 gives the life signal (Chiller OK)

The output signal of Simatic 8A3 drives the relays as follows:

LED	Relay	Description
LED 0	19K1	Life Bit, LED on 19K1 is on
LED1	19K2	Error in water circuit
LED 2	19K3	Error in refrigerant circuit

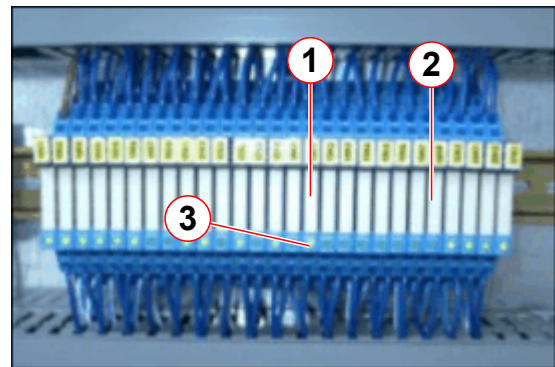
LED 3 to 7 and relay 19K4 to 19K8 are used for error codes.

Fig. 36: Simatic S7



- (1) Digital output 8 A3
- (2) Digital output LED 0

Fig. 37: Relay



- (1) Relay 19K1
- (2) Relay 19K8
- (3) LED of Relay 19K1

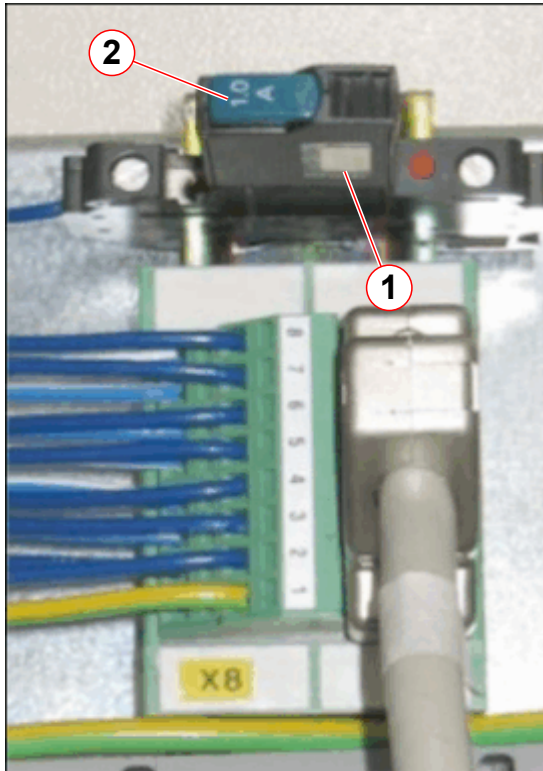
- If the LED 0 of the digital output is on
 - » Chiller has no error
 - » LED on relay 19K1 is on

If the LED on relay 19K1 is off the relay is damaged.

Do a check of the interface cabling and replace the relay.

2.8.1.1 Checking the circuit breaker 28F1

Fig. 38: Circuit breaker



- (1) Circuit breaker 28F1
- (2) Reset button

- Press the blue cap to reset the fuse.

2.8.1.2 Measure the input voltage of the ACS controller

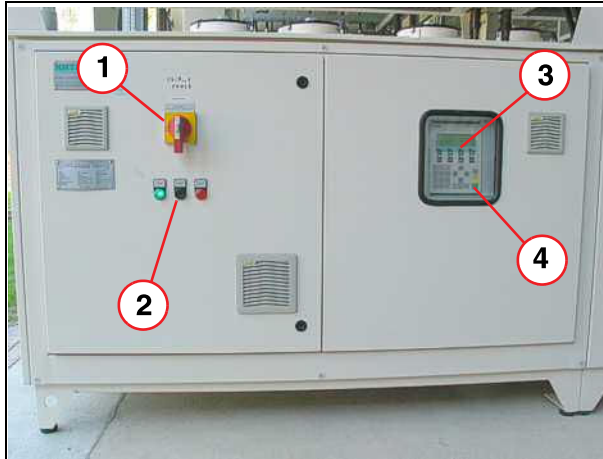
Measure the input voltage from the chiller for the ACS controller if the chiller is ok but the eventlog shows "chiller not ok".

- Measure the voltage between pin1 and pin14 at the D-sub connector X802 in front of the ACS.
 - » If 24 VDC is measured, it is assumed that the ACS control board is defective.

2.8.2 Switch off / on procedure of the Chiller

1. Switch off the chiller. Using ON - OFF switch (→ 1/ Fig. 39 Page 43).

Fig. 39: Chiller



- (1) ON - OFF switch
- (2) Reset switch
- (3) S7 controller
- (4) S7 - ACK button

2. Switch on the chiller. Use the ON - OFF switch (→ 1/ Fig. 39 Page 43).
3. Check the messages on controller S7 (→ 3/ Fig. 39 Page 43).
4. In case an error occurs during the switch-on process, press the **"Reset"** button and the **"ACK"** button on controller S7 (→ 4/ Fig. 39 Page 43).

2.8.3 Automatic restart procedure for the chiller

2.8.3.1 Main switch: On ==> Off ==> On

When switching the main switch off and on again, the operator's display shows the "phase sequence" (flashing) failure message and the green and red indicator lamps on the electrical cabinet also flash to indicate this failure.

After a start-up time of around 20 seconds the chiller will restart automatically and the life bit is on again.

The flashing red indicator lamp goes off again and the green lamp continues flashing to indicate the interruption.

The chiller restarts automatically.

- » In case the chiller does not restart automatically, the failure is still pending.

2.8.3.2 Mains supply: ok ==> power failure ==> power returns

Once power returns after a power failure, the screen displays the "phase sequence" error message and the green indicator lamp at the electrical cabinet flashes to indicate that a failure had occurred.

The chiller restarts automatically.



If the power supply is not within the limit values set at the phase sequence relay (over-voltage exceeds 528 V (440 V), undervoltage remains/falls under 432 V (360 V), and asymmetry is larger than 10%), the error still exists. The red indicator lamp at the electrical cabinet flashes to show that an error exists. The chiller does not restart.

Once the voltage supply has stabilized again, the chiller restarts automatically after a start-up time of around 20 secs, the red indicator lamp goes off, the life bit is on again, the green indicator lamp saying that a failure was pending continues to flash, and the "phase sequence" failure message remains visible at the display.

- » If the chiller does not restart automatically, the failure is still pending or another failure exists!
- » To set off the failure message at the operator's display, press the "ACK-button" at the display unit.
- » The flashing green or red indicator lamp at the electrical cabinet remains permanently on (green lamp) or goes off (red lamp) if the "black RESET-button" at the electrical cabinet is actuated.

2.8.4 Drain procedure of the chiller (water circuit)



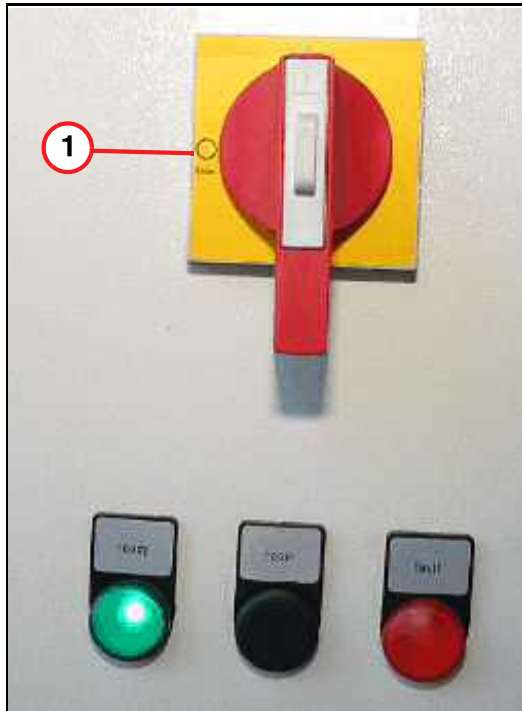
Chilled water containing ethylene glycol should be collected while draining. This water should be refilled into the cooling circuit.

2.8.4.1 Required Tools

- Filling adapter
- Bucket
- Water pipe

2.8.4.2 Procedure

Fig. 40: Chiller

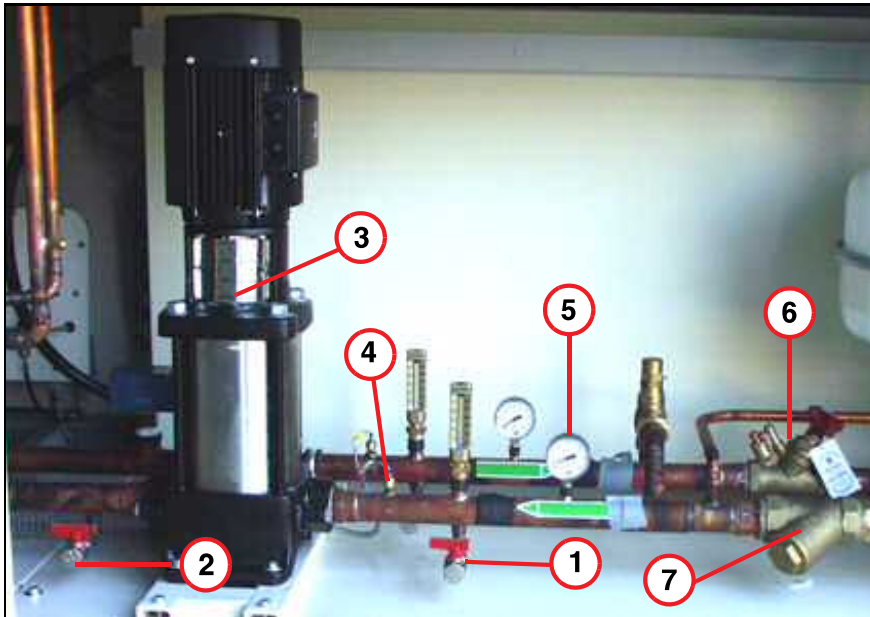


(1) Main Switch 5Q1

- Switch off the chiller using main switch 5Q1 of the chiller

- Close the shut off valves

Fig. 41: Chiller



- (1) Drain and fill valve 1
- (2) Drain and fill valve 2
- (3) Water pump
- (4) Temperature sensor
- (5) Manometer
- (6) Flow regulation valve
- (7) Strainer

Fig. 42: Drain and fill valve



(1) Valve closed

- Connect the water pipe fitted to the adapter to the drain and fill valve 1
- Open the drain valve and drain the Chiller into a bucket

2.8.5 Filling from the chiller



Maximum static filling pressure when the chiller has been switched off is 1.5 bar.



Do not use any connection to a drinking water system!



Filling is possible with running Chiller



The water circuit from the chiller to the MR system needs to be filled with 35% to a maximum of 38% vol. ethylene glycol.

35% to 38% vol. ethylene glycol mixture corresponds to approx. -21 °C to -23 °C (-5.8 °F to -9.4 °F).

Only ethylene glycol may be used.

When starting up the chiller, the chiller start-up personnel should leave 30 liters of ethylene glycol mixture at the system.

This reserve supply can be used for refills.

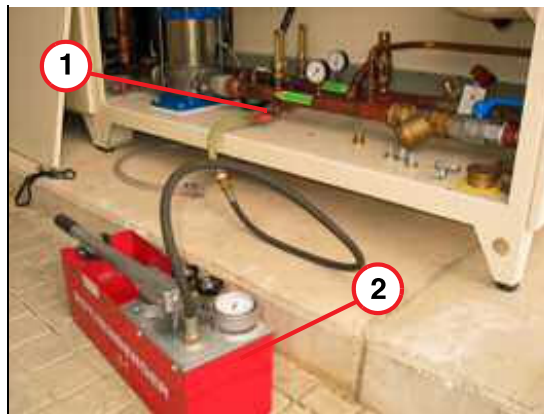
Refill if water pressure return < 0.5 bar or pressure difference is out of specification.

Filling pressure depends on the system configuration:

Tab. 1 Filling pressure

System configuration	Fill pressure at Chiller manometer Chiller is off	Warning level setting: SeSo / Magnet&Cooling / ACS / Water Pressure Return
MR system & Chiller are at the same level and pipe routing is higher than the MR system or Chiller is above the MR system.	0.1 - 0.5 bar (overpressure) + 0.1 bar/m (difference in level)	Warning level = 0.1 bar (overpressure) + 0.01 bar/m (flow resistance) + 0.1 bar/m (difference in level)
Example: Chiller or pipe routing is 10m above the MR system. Length of piping 20m	0.5 bar (overpressure) + 10 m * 0.1 bar/m (difference in level) = 1.5 bar (fill pressure)	Warning level = 0.1 bar + 0.01 bar/m * 20 m + 0.1 bar/m * 10m = 1.3 bar
Example: Chiller and MR system are on the same level. Pipe routing is 5m above the MR system, pipe length is 20m	0.5 bar (overpressure) + 5m * 0.1 bar/m (difference in level) = 1.0 bar (fill pressure)	Warning level = 0.1 bar + 0.01 bar/m * 20 m + 0.1 bar/m * 5m = 0.8 bar
MR system & Chiller & pipe routing are on the same level.	0.1 - 0.5 bar (overpressure)	0.4 bar
MR system is higher than the Chiller.	0.1 - 0.5 bar (overpressure) + 0.1 bar/m (difference in level)	0.4 bar
Example: Chiller is 5 m below the MR system.	0.5 bar (overpressure) + 5m * 0.1 bar/m (difference in level) = 1.0 bar (fill pressure)	0.4 bar

Fig. 43: Filling the chiller



- (1) Fill valve
- (2) Hand pump

- Connect the water hose to the filling valve

Fig. 44: Filling the hand pump



- Fill the pump with ethylene glycol water mixture

- Open the fill valve of the chiller.

Fig. 45: Pressure manometer



(1) Manometer

- Stop filling when the pressure manometer shows the correct value.

- Open the SESO platform "Magnet&Cooling"
Calibrate the flow sensors.
Check pressure of the system via SeSo

Magnet & Cooling / Status	
■ Check water pressure, supply ■ Check water pressure, return ■ Calculate the mathematical difference between water pressure supply and water pressure return	
» The mathematical difference (water pressure supply minus water pressure return) must range between 4.0 bar and 4.5 bar for 50 and 60 Hz.	
» Refill if water pressure return shows <0.5 bar or pressure difference is out of specification.	
Warning level for pressure return is at 0.4 bar.	

Warning level system with IFP Values are measured in the ACS.	
Water pressure, supply max.	6.0 bar
Water pressure, return min.	0.4 bar
Water pressure, return max.	2.8 bar
Water temperature, supply max.	25

2.8.6 Bleeding the Chiller water pump

i

It is possible that air bubbles in the water cooling circuit collect in the water pump of the Chiller.

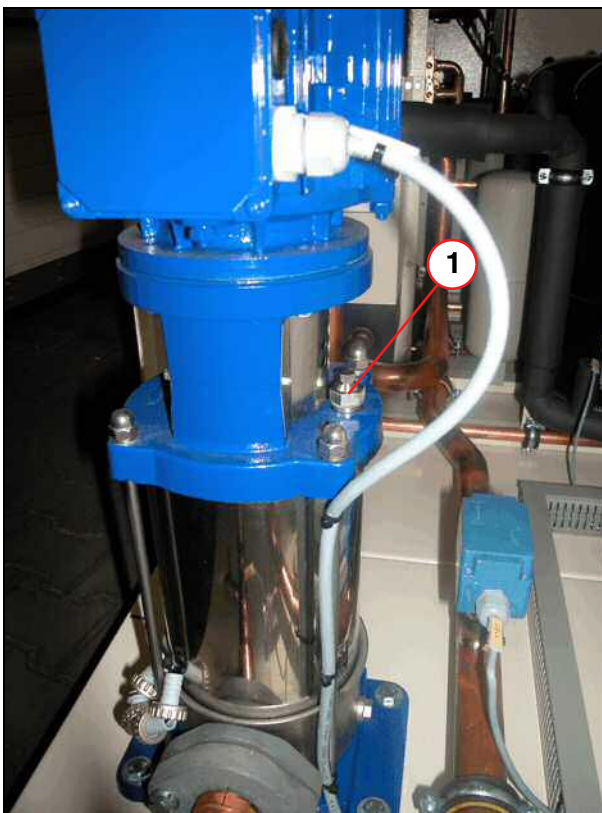
The bubbles prevent sufficient cooling of the water pump.

This will damage the slide ring seal.

As a remedy, bleed the pump after operating it for a few days.

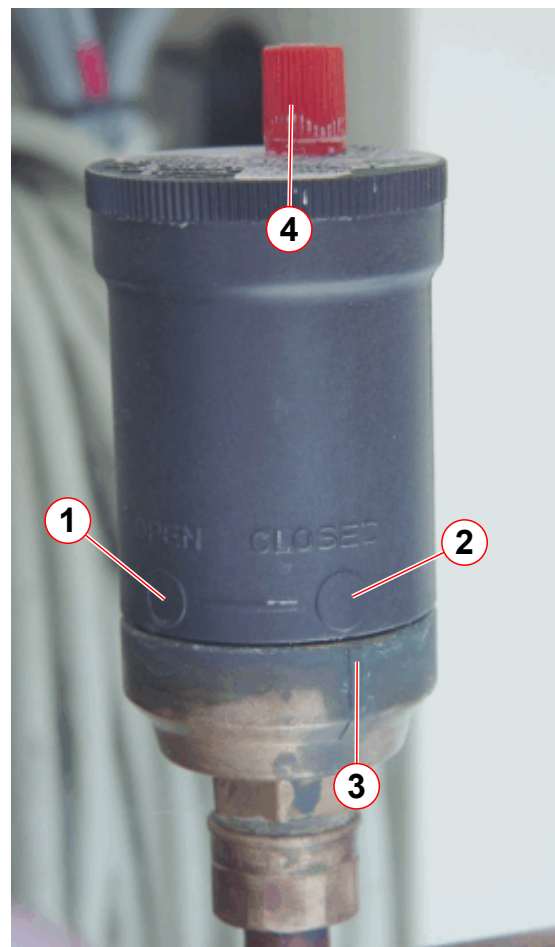
2.8.6.1 Bleeding valve (old version, Chiller serial number <01195)

Fig. 46: Water pump



(1) Bleeding valve

Fig. 47: Bleeding valve



- (1) Position "OPEN"
- (2) Position "CLOSED"
- (3) Reference line
- (4) Cap

- Open the bleeding valve of the pump.(→ 1/ Fig. 46 Page 51)
 - » Leave it open until air escapes.

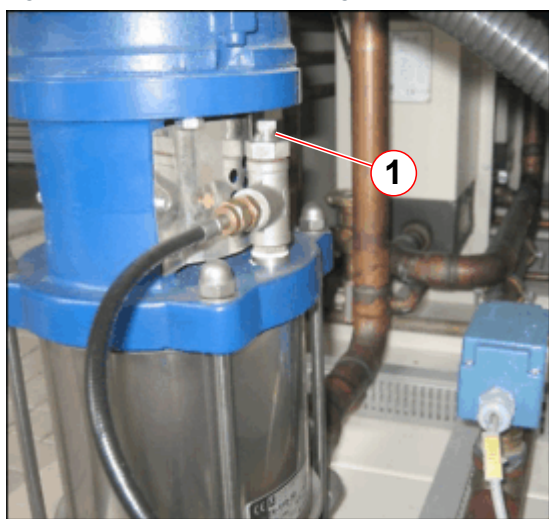
- For bleeding the water circuit open the bleeding valve : (→ Fig. 47 Page 51)
The bleeding valve is located in the cabinet left to the water pump.
 - » Turn the valve to "OPEN" position, check if the cap (→ 4/ Fig. 47 Page 51) is loosen.
Leave the valve in "OPEN" position for some hours with chiller is switch on.



Do not forget turning the valve to "CLOSED" position.
Keep the cap loosen. Do not remove the cap it.

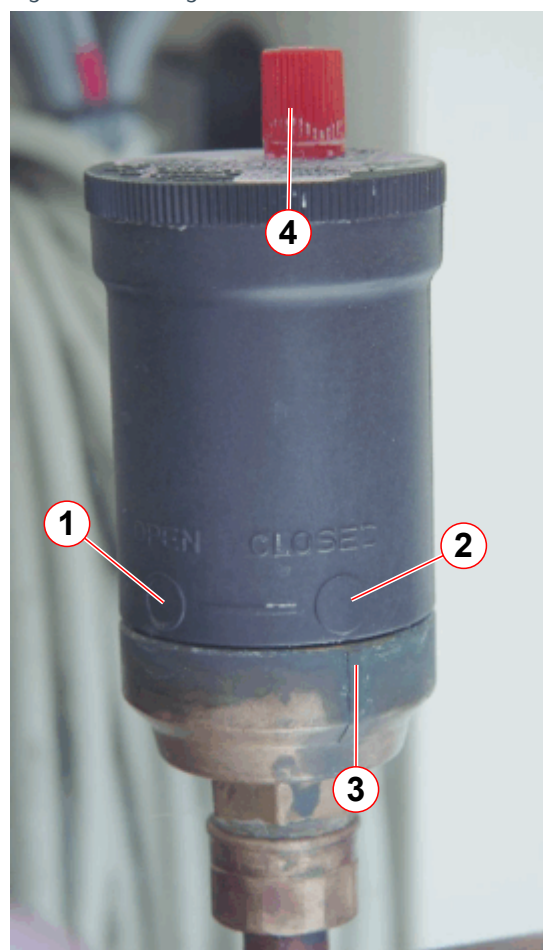
2.8.6.2 Self-controlled bleeding valve (version 1)

Fig. 48: self-controlled bleeding valve (Version 1)



(1) Bleeding valve

Fig. 49: Bleeding valve



- (1) Position "OPEN"
- (2) Position "CLOSED"
- (3) Reference line
- (4) Cap

- Open the bleeding valve of the pump. (→ 1/ Fig. 48 Page 52)
 - » Leave it open until air escapes.

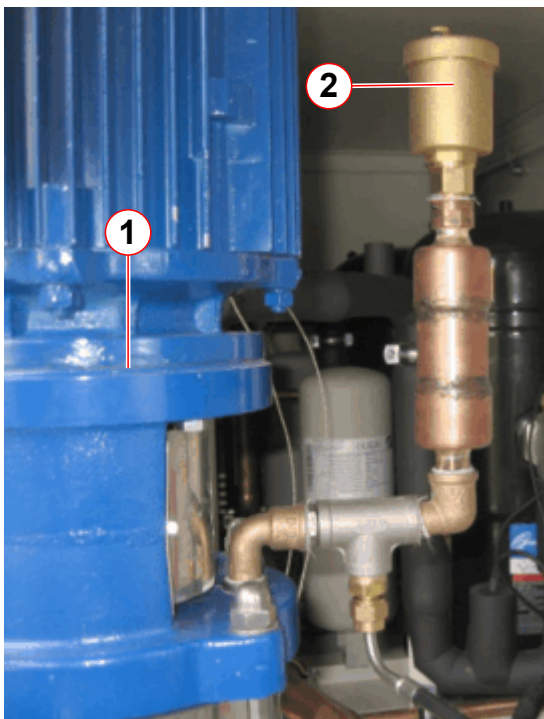
- For bleeding the water circuit open the bleeding valve (→ 1, 2/ Fig. 49 Page 52):
The bleeding valve is located in the cabinet left to the water pump.
 - » Turn the valve to "OPEN" position, check if the cap (→ 4/ Fig. 49 Page 52) is loosened. Leave the valve in "OPEN" position for some hours with chiller is switch on.



Do not forget turning the valve to "CLOSED" position.
Keep the cap loosened. Do not remove it.

2.8.6.3 Self-controlled bleeding valve (Version 2, serial number >1696)

Fig. 50: Self-controlled bleeding valve (Version 2)



- (1) Water pump
- (2) Bleeding valve

Fig. 51: Self-controlled bleeding valve (Version 2)



- (1) Water pump
- (2) Bleeding valve
- (3) Cap

For bleeding the water circuit loosen the cap of the bleeding valve.(→ 3/ Fig. 51 Page 53)

- » Leave the cap loosened.

2.9 Testing the ethylene glycol level



The specification is 35% vol. to a maximum of 38% vol. of ethylene glycol. Do not use propylene glycol or automotive antifreeze!

The chiller Service must do a check of this concentration during chiller maintenance with a special measurement device.

Antifreeze tester are specified only for certain glycol brands.

2.9.1 Required Tools

- A special antifreeze tester. Contact the chiller service.

2.9.2 Procedure

- Open the drain valve on the ACS or on the chiller and drain about 0.5 l ethylene glycol mixture into a bucket.
- Via the temperature scale, do a check of the ethylene glycol level with the ethylene glycol tester.
 - » -21 °C to -23 °C (-5.8 °F to -9.4 °F) corresponds to approx. 35% to 38% vol. of ethylene glycol water mixture.
- Increase the ethylene glycol level, if necessary.



Do not mix different types of glycol with one another, since this can result in flocculation and problems in the chiller.

If an incorrect type of glycol was added, the water circuit needs to be drained completely. Refill the water circuit with a 35% vol. to a maximum of 38% vol. of ethylene glycol water mixture.

Please contact the chiller start-up personnel.

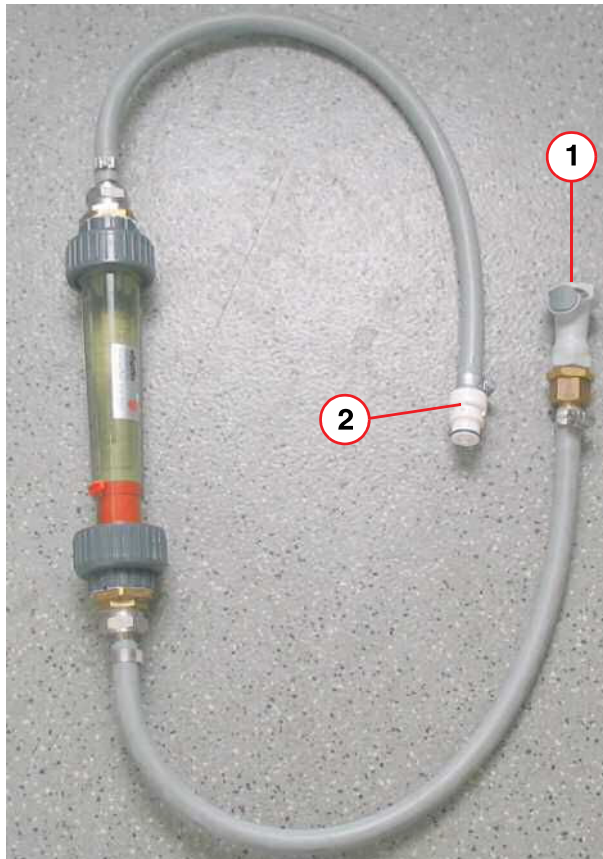
If the chiller is operated without antifreeze, the chiller can be damaged.

2.10 Testing the water flow to the RFPA and the gradient power stage

2.10.1 Required tool

Water flow meter 10018564

Fig. 52: Water flow meter GPA / RFPA



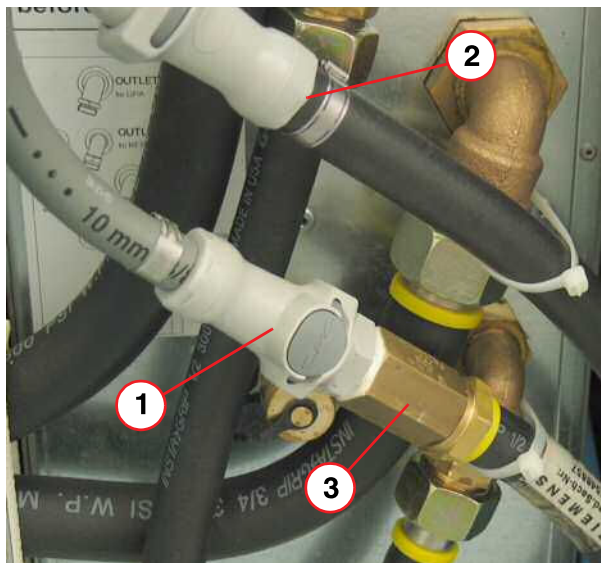
- (1) Quick connector (female)
- (2) Quick connector (male)

2.10.2 Test procedure of the RFPA

- Switch off the water pump in the SEP
or
Switch off the Chiller (IFP configuration).

- Disconnect the water pipe "outlet to RFPA" at the ACS.(→ 1/ Fig. 53 Page 56)

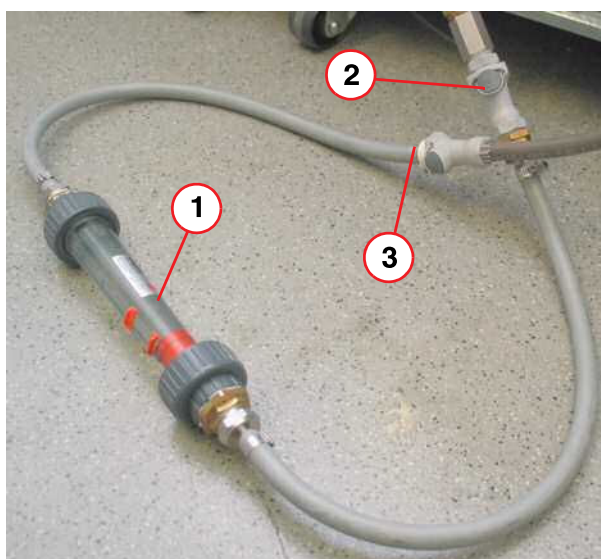
Fig. 53: ACS - RFPA



- (1) Outlet to RFPA
- (2) Inlet from RFPA
- (3) Flow limiter

- Connect the flow meter

Fig. 54: Water flow meter



- (1) Water flow meter
- (2) Water in (female)
- (3) Water out (male)



Do a check of the correct water flow direction through the flow meter.

- Switch on the water pump or the chiller.

- Do a check of the flow rate

Fig. 55: Water flow meter



- (1) Water in
(2) Water out

	Nominal value
Avanto (DORA)	8 - 10 l/min
Esprea (PORA)	10 - 16 l/min
Trio (Dressler)	8 - 10 l/min

- Switch off the water pump or the chiller before disconnecting the flow meter.

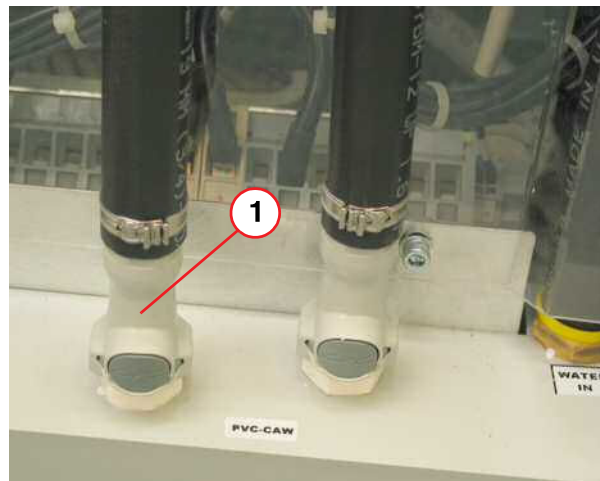
2.10.3 Test procedure for the gradient power amplifier

- Switch off the water pump in the SEP
or
Switch off the Chiller (IFP configuration).
- Disconnect the water pipe at the gradient power amplifier.

Fig. 56: GPA



Fig. 57: GPA



(1) Quick connector

- Connect the flow meter.



Do a check of the correct water flow direction through the flow meter.

- Switch on the water pump or the chiller.
- Do a check of the flow rate.
 - » Nominal flow rate: 15 l/min +1/ -2 l/min
- Switch off the water pump in the SEP or the chiller (IFP configuration) before disconnecting the flow meter.

2.11 Conversion list of temperature sensors

Temperature (°C)	PT 100 (0.4 ohms/°C)	KTY (~15 ohms/°C)
	Location: SEP: R3 ACS: R1, R2, R3, R4, R5 IFP: Flow sensor, temperature sensor	Location: SEP: R1, R2 (temperature sensor)
0	100 ohms	1,634 ohms
25	110 ohms	2,000 ohms +/- 10 ohms
10	104 ohms	1,774 ohms
20	108 ohms	1,922 ohms
30	112 ohms	2,078 ohms
40	116 ohms	2,040 ohms
50	120 ohms	2,410 ohms

- » The common print number (M6-020.840.01) from the Troubleshooting Guide is divided into separate Component print numbers.

Revision 02:

- Chapter 01:
 - Ethylene glycol water mixture
- Chapter 02
 - Ethylene glycol water mixture
 - Following sections removed:
 - Drain procedure for the chiller
 - Filling from the chiller
 - Bleeding the chiller water pump
 - Automatic restart procedure added

Revision 03:

- Chapter 02
 - Testing the ethylene glycol water mixture
 - Bleeding of the chiller updated
 - Chiller interface device updated
 - Conversion list added.

Revision 04:

- Chapter 02
 - Testing the expansion vessel.
 - IFP updated.
 - Automatic restart procedure of the chiller added.

Revision 05:

- Chapter 01
 - Link to "Cooling Error messages" has been added.

Revision 06:

- Chapter (→ Procedures / Page 9)
 - Update of the water pressure for 50 Hz / 60 Hz

Revision 07:

- Chapter (→ Procedures / Page 9)
 - Update of chapter "Filling from the SEP"

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Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthineers

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